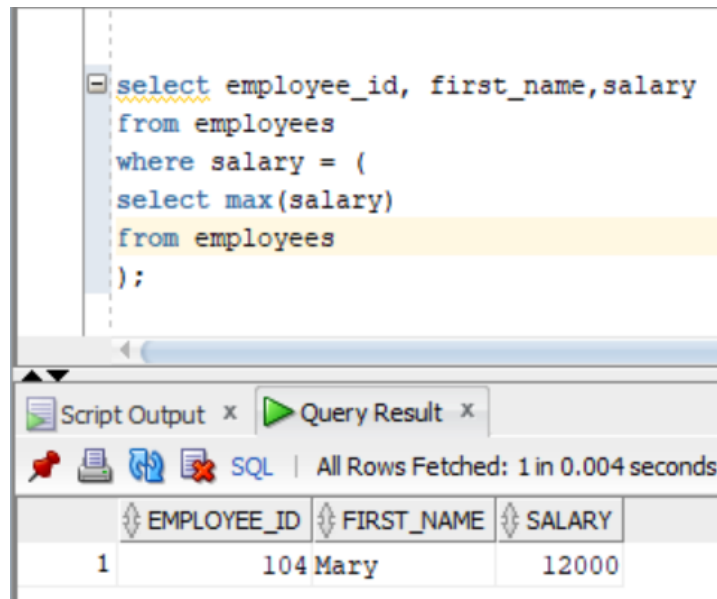


DB Lab 06 Submission

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Roll Number: 24k-0737

Task 01:



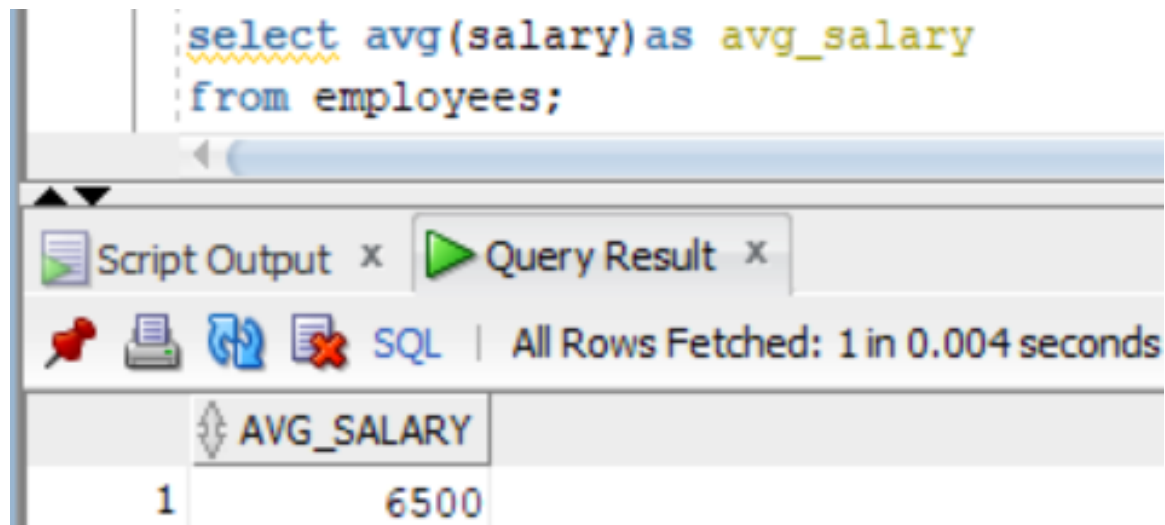
The screenshot shows a SQL query editor with the following query:

```
select employee_id, first_name, salary
from employees
where salary = (
select max(salary)
from employees
);
```

Below the query editor, the 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0.004 seconds'.

	EMPLOYEE_ID	FIRST_NAME	SALARY
1	104	Mary	12000

Task 02:



The screenshot shows a SQL query editor with the following query:

```
select avg(salary) as avg_salary
from employees;
```

Below the query editor, the 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0.004 seconds'.

	AVG_SALARY
1	6500

<pre> select employee_id, first_name, salary from employees where salary > (select avg(salary) from employees); </pre>			
Script Output x Query Result x SQL All Rows Fetched: 5 in 0.005 seconds			
	EMPLOYEE_ID	FIRST_NAME	SALARY
1	104	Mary	12000
2	101	John	7000
3	103	Ali	8000
4	105	Ahmed	9000
5	109	Sofia	9500

Task 03:

<pre> select employee_id, first_name, salary from employees where salary = All (select min(salary) from employees); </pre>			
Script Output x Query Result x SQL All Rows Fetched: 1 in 0.006 seconds			
	EMPLOYEE_ID	FIRST_NAME	SALARY
1	102	Sara	3000

Task 04:

Worksheet

Query Builder

```
select employee_id, first_name, salary
from employees
where salary = (
select avg(salary)
from departments
where department_id = 20
);
```

Script Output x

Query Result x

 SQL | All Rows Fetched: 10 in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	SALARY
1	104	Mary	12000
2	107	David	5000
3	101	John	7000
4	102	Sara	3000
5	103	Ali	8000
6	105	Ahmed	9000
7	106	Nina	4000
8	108	Alex	3500
9	109	Sofia	9500
10	110	James	4000

Task 05:

```
select hire_date, first_name
from employees;
```

Script Output x Query Result x

SQL | All Rows Fetched: 10 in

	HIRE_DATE	FIRST_NAME
1	28-FEB-22	Mary
2	05-JAN-22	David
3	10-JAN-22	John
4	15-MAR-22	Sara
5	20-MAY-22	Ali
6	10-APR-22	Ahmed
7	12-JUN-22	Nina
8	01-JUL-22	Alex
9	20-AUG-22	Sofia
10	15-SEP-22	James

```
select first_name, hire_date
from employees
where hire_date = (
select max(hire_date)
from employees
);
```

Script Output x Query Result x

SQL | All Rows Fetched: 1 in 0.007 seconds

	FIRST_NAME	HIRE_DATE
1	James	15-SEP-22

Task 06:

```
select *
from employees e
where e.department_id in (
select department_id
from departments
where location_id = 1700
);
```

Script Output x Query Result x

SQL | All Rows Fetched: 6 in 0.017 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	MANAGER_ID	DEPARTMENT_ID
1	101 John	Smith	john.smith@company.com	100-1111	10-JAN-22	IT_PROG	7000	104	60
2	102 Sara	Jones	sara.jones@company.com	100-2222	15-MAR-22	SH_CLERK	3000	101	50
3	103 Ali	Khan	ali.khan@company.com	100-3333	20-MAY-22	SA_REP	8000	104	80
4	105 Ahmed	Ali	ahmed.ali@company.com	100-5555	10-APR-22	IT_PROG	9000	101	60
5	109 Sofia	Martinez	sofia.martinez@company.com	100-9999	20-AUG-22	SA_REP	9500	103	80
6	110 James	Wilson	james.wilson@company.com	100-1010	15-SEP-22	SH_CLERK	4000	102	50

Task 07:

```
select *
from employees e
where e.job_id = (
select job_id
from departments
where department_id = 80 );
```

Script Output x Query Result x

SQL | All Rows Fetched: 10 in 0.008 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	MANAGER_ID	DEPARTMENT_ID
1	104 Mary	Brown	mary.brown@company.com	100-4444	28-FEB-22	MK_MAN	12000	(null)	20
2	107 David	Lee	david.lee@company.com	100-7777	05-JAN-22	HR_REP	5000	(null)	40
3	101 John	Smith	john.smith@company.com	100-1111	10-JAN-22	IT_PROG	7000	104	60
4	102 Sara	Jones	sara.jones@company.com	100-2222	15-MAR-22	SH_CLERK	3000	101	50
5	103 Ali	Khan	ali.khan@company.com	100-3333	20-MAY-22	SA_REP	8000	104	80
6	105 Ahmed	Ali	ahmed.ali@company.com	100-5555	10-APR-22	IT_PROG	9000	101	60
7	106 Nina	Patel	nina.patel@company.com	100-6666	12-JUN-22	HR_REP	4000	107	40
8	108 Alex	Taylor	alex.taylor@company.com	100-8888	01-JUL-22	AD_ASST	3500	104	10
9	109 Sofia	Martinez	sofia.martinez@company.com	100-9999	20-AUG-22	SA_REP	9500	103	80
10	110 James	Wilson	james.wilson@company.com	100-1010	15-SEP-22	SH_CLERK	4000	102	50

Task 08:

```
select *
from employees
where salary > ALL (
select salary
from employees
where job_id = 'SH_CLERK' );
```

Script Output x Query Result x

SQL | All Rows Fetched: 6 in 0.004 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	MANAGER_ID	DEPARTMENT_ID
1	107 David	Lee	david.lee@company.com	100-7777	05-JAN-22	HR_REP	5000	(null)	40
2	101 John	Smith	john.smith@company.com	100-1111	10-JAN-22	IT_PROG	7000	104	60
3	103 Ali	Khan	ali.khan@company.com	100-3333	20-MAY-22	SA_REP	8000	104	80
4	105 Ahmed	Ali	ahmed.ali@company.com	100-5555	10-APR-22	IT_PROG	9000	101	60
5	109 Sofia	Martinez	sofia.martinez@company.com	100-9999	20-AUG-22	SA_REP	9500	103	80
6	104 Mary	Brown	mary.brown@company.com	100-4444	28-FEB-22	MK_MAN	12000	(null)	20

Task 09:

```
select *
from employees
where salary < ANY (
select salary
from employees
where job_id = 'IT_PROG' );
```

Script Output x Query Result x

SQL | All Rows Fetched: 7 in 0.008 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	MANAGER_ID	DEPARTMENT_ID
1	102 Sara	Jones	sara.jones@company.com	100-2222	15-MAR-22	SH_CLERK	3000	101	50
2	108 Alex	Taylor	alex.taylor@company.com	100-8888	01-JUL-22	AD_ASST	3500	104	10
3	110 James	Wilson	james.wilson@company.com	100-1010	15-SEP-22	SH_CLERK	4000	102	50
4	106 Nina	Patel	nina.patel@company.com	100-6666	12-JUN-22	HR_REP	4000	107	40
5	107 David	Lee	david.lee@company.com	100-7777	05-JAN-22	HR_REP	5000	(null)	40
6	101 John	Smith	john.smith@company.com	100-1111	10-JAN-22	IT_PROG	7000	104	60
7	103 Ali	Khan	ali.khan@company.com	100-3333	20-MAY-22	SA_REP	8000	104	80

Task 10:

```
select *
from employees e
where e.department_id in (
select department_id
from departments
where department_name like '%Sales%');
```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.005 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	MANAGER_ID	DEPARTMENT_ID
1	103 Ali	Khan	ali.khan@company.com	100-3333	20-MAY-22	SA_REP	8000	104	80
2	109 Sofia	Martinez	sofia.martinez@company.com	100-9999	20-AUG-22	SA_REP	9500	103	80

Task 11:

```
select *
from employees e
where salary > (
select avg(salary)
from employees d
where e.department_id = d.department_id
);
```

Script Output x Query Result x

SQL | All Rows Fetched: 4 in 0.015 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	MANAGER_ID	DEPARTMENT_ID
1	107 David	Lee	david.lee@company.com	100-7777	05-JAN-22	HR_REP	5000	(null)	40
2	105 Ahmed	Ali	ahmed.ali@company.com	100-5555	10-APR-22	IT_PROG	9000	101	60
3	109 Sofia	Martinez	sofia.martinez@company.com	100-9999	20-AUG-22	SA_REP	9500	103	80
4	110 James	Wilson	james.wilson@company.com	100-1010	15-SEP-22	SH_CLERK	4000	102	50

Task 12:

Worksheet Query Builder

```
select *  
from employees e  
where salary = (  
select max(salary)  
from employees d  
where e.department_id = d.department_id  
);
```

Script Output x Query Result x

SQL | All Rows Fetched: 6 in 0.003 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER
1	104	Mary	Brown	mary.brown@company.com	100-4444
2	107	David	Lee	david.lee@company.com	100-7777
3	105	Ahmed	Ali	ahmed.ali@company.com	100-5555
4	108	Alex	Taylor	alex.taylor@company.com	100-8888
5	109	Sofia	Martinez	sofia.martinez@company.com	100-9999
6	110	James	Wilson	james.wilson@company.com	100-1010

Task 13:

```
select *  
from employees e  
where salary > (  
select avg(salary)  
from employees d  
where e.job_id = d.job_id  
);
```

Script Output x Query Result x

SQL | All Rows Fetched: 4 in 0.007 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER
1	107	David	Lee	david.lee@company.com	100-7777
2	105	Ahmed	Ali	ahmed.ali@company.com	100-5555
3	109	Sofia	Martinez	sofia.martinez@company.com	100-9999
4	110	James	Wilson	james.wilson@company.com	100-1010

Task 14:

```
select employee_id , first_name,hire_date,department_id
from employees e
where hire_date > (
select min(hire_date)
from employees d
where e.department_id = d.department_id);
```

Script Output x Query Result x

SQL | All Rows Fetched: 4 in 0.003 seconds

	EMPLOYEE_ID	FIRST_NAME	HIRE_DATE	DEPARTMENT_ID
1	105	Ahmed	10-APR-22	60
2	106	Nina	12-JUN-22	40
3	109	Sofia	20-AUG-22	80
4	110	James	15-SEP-22	50

Task 15:

```
select *
from employees e
where e.manager_id in (
select d.manager_id
from employees d
where e.manager_id = d.manager_id);
```

Script Output x Query Result x

SQL | All Rows Fetched: 8 in 0.002 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	MANAGER_ID	DEPARTMENT_ID
1	101	John	Smith	john.smith@company.com	100-1111	10-JAN-22	IT_PROG	7000	104	60
2	103	Ali	Khan	ali.khan@company.com	100-3333	20-MAY-22	SA_REP	8000	104	80
3	108	Alex	Taylor	alex.taylor@company.com	100-8888	01-JUL-22	AD_ASST	3500	104	10
4	102	Sara	Jones	sara.jones@company.com	100-2222	15-MAR-22	SH_CLERK	3000	101	50
5	105	Ahmed	Ali	ahmed.ali@company.com	100-5555	10-APR-22	IT_PROG	9000	101	60
6	106	Nina	Patel	nina.patel@company.com	100-6666	12-JUN-22	HR_REP	4000	107	40
7	109	Sofia	Martinez	sofia.martinez@company.com	100-9999	20-AUG-22	SA_REP	9500	103	80
8	110	James	Wilson	james.wilson@company.com	100-1010	15-SEP-22	SH_CLERK	4000	102	50

Task 16:

```
select *  
from employees e  
where e.manager_id NOT IN (  
select d.manager_id  
from employees d  
where e.manager_id = d.manager_id);
```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.005 seconds

	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	MANAGER_ID
1 n@company.com	100-4444	28-FEB-22	MK_MAN	12000	(null)
2 @company.com	100-7777	05-JAN-22	HR_REP	5000	(null)

Task 17:

```
select *  
from departments  
where department_id NOT IN (  
select department_id  
from employees );
```

Script Output x Query Result (x)

SQL | All Rows Fetched: 3 in 0.004 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	LOCATION_ID
1	30	Purchasing	2000
2	70	Public Relations	3000
3	90	Executive	1000

Task 18:

```
select *  
from departments  
where department_id = ANY (  
select department_id  
from employees  
where salary > 10000);
```

Script Output x Query Result x

SQL | All Rows Fetched: 1 in 0.005 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	LOCATION_ID
1	20	Marketing	2000

Task 19:

```
SELECT *  
FROM Departments d  
WHERE department_id IN (  
SELECT department_id  
FROM Employees  
GROUP BY department_id  
HAVING AVG(salary) > 8000  
);
```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.008 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	LOCATION_ID
1	20	Marketing	2000
2	80	Sales	1700

Task 20:

```
SELECT *  
FROM (  
    SELECT d.department_id, d.department_name, AVG(e.salary) AS avg_salary  
    FROM Departments d  
    JOIN Employees e ON d.department_id = e.department_id  
    GROUP BY d.department_id, d.department_name  
    ORDER BY AVG(e.salary) DESC  
)  
WHERE ROWNUM <= 3;
```

Script Output x Query Result x

SQL | All Rows Fetched: 3 in 0.007 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	AVG_SALARY
1	20	Marketing	12000
2	80	Sales	8750
3	60	IT	8000